WEB PROGRAMMING MINI PROJECT

Course: Fundamentals of Computing (FOC)

Subject code-GXEST203

submitted by

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To

the APJ Abdul kalam Technological University in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology

in

Computer Science and Engineering

Department of Computer Science and Engineering

College of Engineering Muttathara

Thiruvananthapuram

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1]

Q1: Create a simple div with an id "box". Add some text content inside the div. Set its background color to blue.

Q2: Create 3 headings with h1, h2 & h3. Give them all a class "heading" & set color of "heading" to red.

Q3:Create a button & set its background color to :

1. green using css stylesheet
2. blue using <style> tag
3.pink using inline code

OUTPUT

HELLO ALL

BTECH

ENGINEERING

CS

Green Blue Pink

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" href="project1.css">
    <style>
        #box { background: blue; color: white; }
        .heading { color: red; }
        .int { background: blue; }
    </style>
</head>
<body>

<div id="box">HELLO ALL</div>

<h1 class="heading">BTECH</h1>
<h2 class="heading">ENGINEERING</h2>
<h3 class="heading">CS</h3>

<button class="ext">Green</button>
<button class="int">Blue</button>
<button style="background: pink;">Pink</button>

</body>
</html>
```

html

css

```
# project1.css > .ext
.ext { background: green; }
```

2] Q1: Create a heading centered on the page with all of its text capitalized by default.

Q2: Set the font family of all the content in the document to "Times New Roman".

Q3: Create one div inside another div. Set id & text "outer" for the first one & "inner" for the second one. Set the outer div text size to 25px & inner div text size to 10px.

SOURCE CODE

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" href="project2.css">
</head>
<body>

<h1>Web development</h1>

<div id="outer">
    outer
    <div id="inner">inner</div>
</div>

</body>
</html>
```

OUTPUT

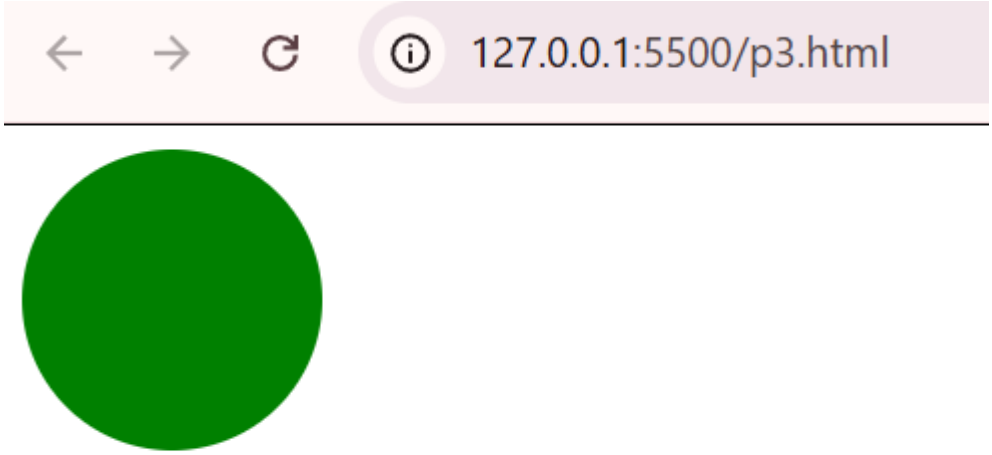
WEB DEVELOPMENT

outer

inner

```
1   body { font-family: "Times New Roman"; }
2
3   h1 {
4       text-align: center;
5       text-transform: uppercase;
6   }
7
8   #outer { font-size: 25px; }
9
10  #inner { font-size: 10px; }
```

OUTPUT



3] Q1: Create a div with height & width of 100px. Set its background color to green & the border radius to 50%.

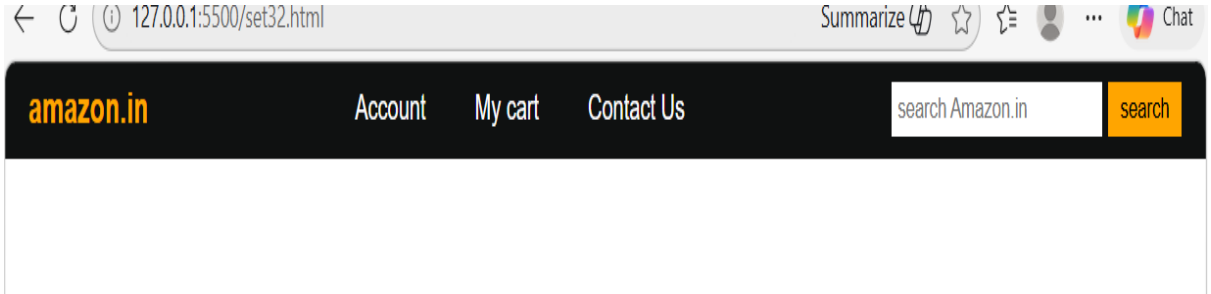
Q2: Create the following navbar.

Q1.

```
p3.html > html > head > link
<!DOCTYPE html>
<html>
<head>
    <title>Q1</title>
    <link rel="stylesheet" href="p3.css">
</head>
<body>

<div class="circle"></div>

</body>
</html>
```

```
.circle {
    width: 100px;
    height: 100px;
    background-color: green;
    border-radius: 50%;
}
```

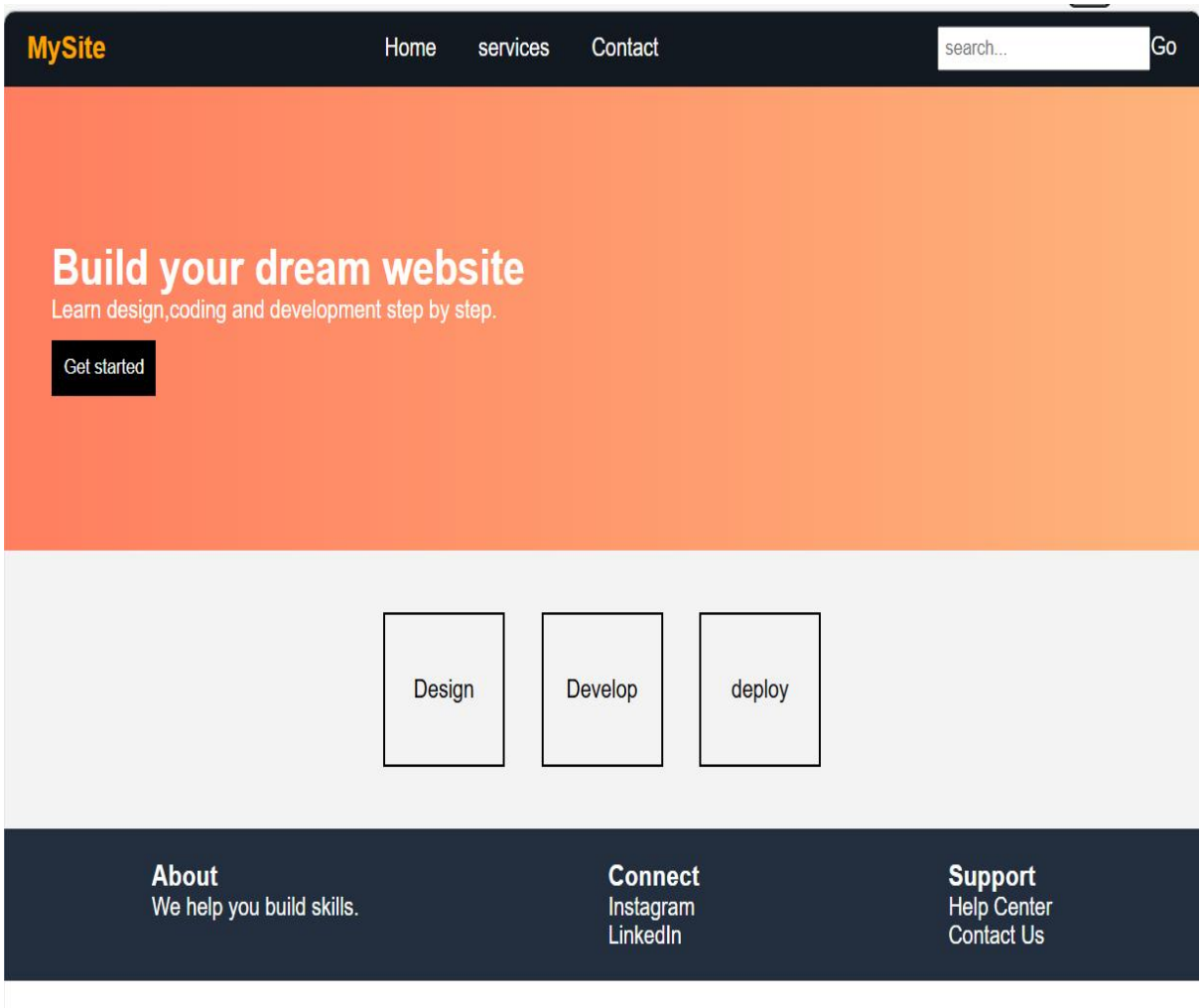
OUTPUT


127.0.0.1:5500/set32.html
Summarize
Chat
amazon.in
Account
My cart
Contact Us
search Amazon.in
search

Q2

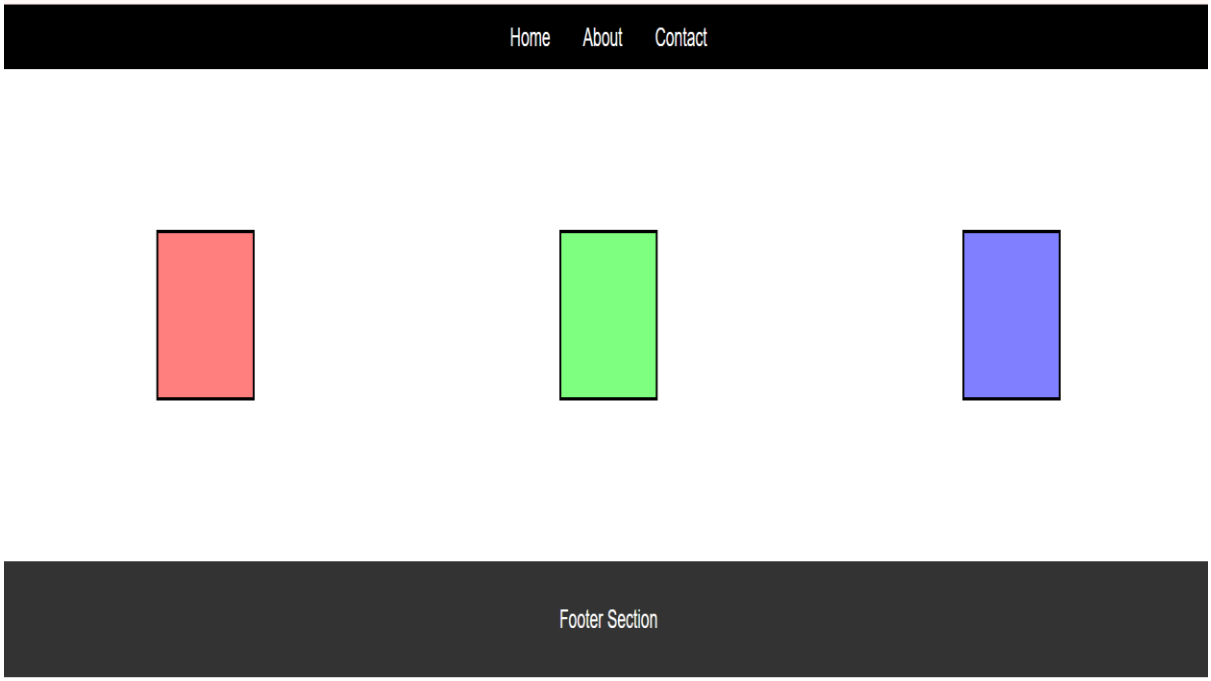
HTML

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="navbar.css">
</head>
<body>
    <nav class="navbar">
        <div class="logo">amazon.in</div>
        <ul class="nav-links">
            <li>Account</li>
            <li>My cart</li>
            <li>Contact Us</li>
        </ul>
        <div class="search-box">
            <input type="text"
            placeholder="search Amazon.in">
            <button>search</button>
        </div>
    </nav>
</body>
</html>
```


MySite
Home
services
Contact
search...
Go
Build your dream website
Learn design,coding and development step by step.
Get started
Design
Develop
deploy
About
We help you build skills.
Connect
Instagram
LinkedIn
Support
Help Center
Contact Us

```
body{
    font-family: Arial, sans-serif;
}
.navbar{
    background-color: #0f1111;
    color: white;
    display: flex;
    align-items: center;
    justify-content: space-between;
    padding: 10px 20px;
}
.logo{
    color: orange;
    font-size: 20px;
    font-weight: bold;
}
.nav-links{
    list-style-type: none;
    display: flex;
    gap: 40px;
    margin: 0;
    padding: 0;
}
.nav-links li{
    cursor: pointer;
}
.search-box input{
    padding: 5px;
    border: none;
    outline: none;
}
.search-box button{
    background-color: orange;
    border: none;
    padding: 5px 10px;
    cursor: pointer;
}
```

OUTPUT



4] Q1: Create a webpage layout with a header, a footer & a content area containing 3 divs. Set the height & width of divs to 100px. (add the previous navbar in the header)

Q2: Add borders to all the divs.

Q3: Add a different background color to each div with an opacity of 0.5.

Q4: Give the content area an appropriate height

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Layout Page</title>
    <link rel="stylesheet" href="p4.css">
</head>
<body>

    <header>
        <nav class="navbar">
            <a href="#">Home</a>
            <a href="#">About</a>
            <a href="#">Contact</a>
        </nav>
    </header>

    <main class="content">
        <div class="box box1"></div>
        <div class="box box2"></div>
        <div class="box box3"></div>
    </main>

    <footer>
        <p>Footer Section</p>
    </footer>

</body>
</html>
```

```css
body {
    margin: 0;
    font-family: Arial, sans-serif;
}
.navbar {
    background-color: black;
    padding: 10px;
    text-align: center;
}
.navbar a {
    color: white;
    margin: 0 15px;
    text-decoration: none;
}
header {
    background-color: #333;
}
.content {
    height: 300px;
    display: flex;
    justify-content: space-around;
    align-items: center;
}
.box {
    width: 100px;
    height: 100px;
    border: 2px solid black;
}
```

```css
.box1 {
    background-color: rgba(255, 0, 0, 0.5);
}

.box2 {
    background-color: rgba(0, 255, 0, 0.5);
}
.box3 {
    background-color: rgba(0, 0, 255, 0.5);
}
footer {
    background-color: #333;
    color: white;
    text-align: center;
    padding: 10px;
}
```

Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit.

Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit.

5] Q1: Create the following layout using the given html. o Give the div a height, width & some background image. o Use the appropriate position property for the div element to place it at the right end of the page. (The div should not move even on scroll) o Use z-index to place the div on top of page.

lorem*5

Love Nature

lorem*5

```html
<!DOCTYPE html>
<html>
<head>
    <title>Layout</title>
    <link rel="stylesheet" href="p5.css">
</head>
<body>

<p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit.

<div class="box">Love Nature</div>

<p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Lorem ipsum dolor sit amet consectetur adipisicing elit.

</body>
</html>
```

```
body {
  <element class="box">        ust to enable scroll */
  Selector Specificity: (0, 1, 0)
.box {
    height: 150px;
    width: 200px;
    background-image: url("https://images.unsplash.com/photo-1501785888041-af3ef285b470");
    background-size: cover;
    color: white;
    font-size: 20px;
    text-align: center;
    line-height: 150px;

    position: fixed;
    right: 0;
    top: 50px;

    z-index: 999;
}
```

6] Q1: Create a navbar with 4 options in the form of anchor tags inside list items. Now, use flexbox to place them all spaced equally in a single line.

Q2: Use flexbox to center one div inside another div.

Q3: Which has higher priority - align-items or align-self?

Answer: align-self has higher priority than align-items.

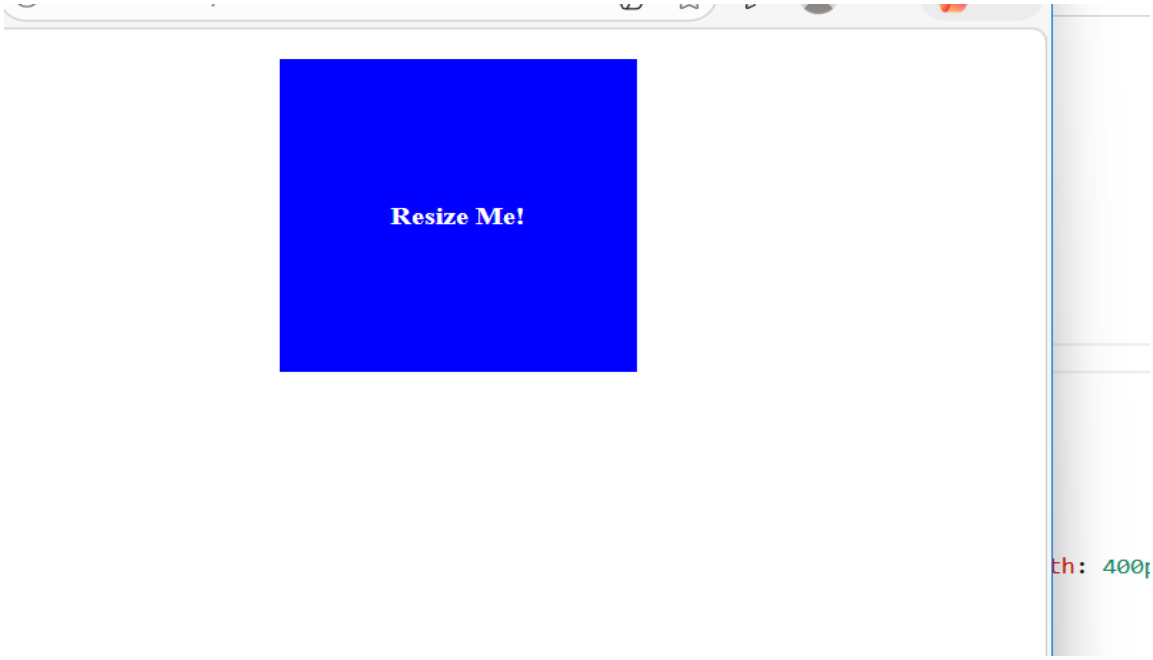
Home About Services Contact

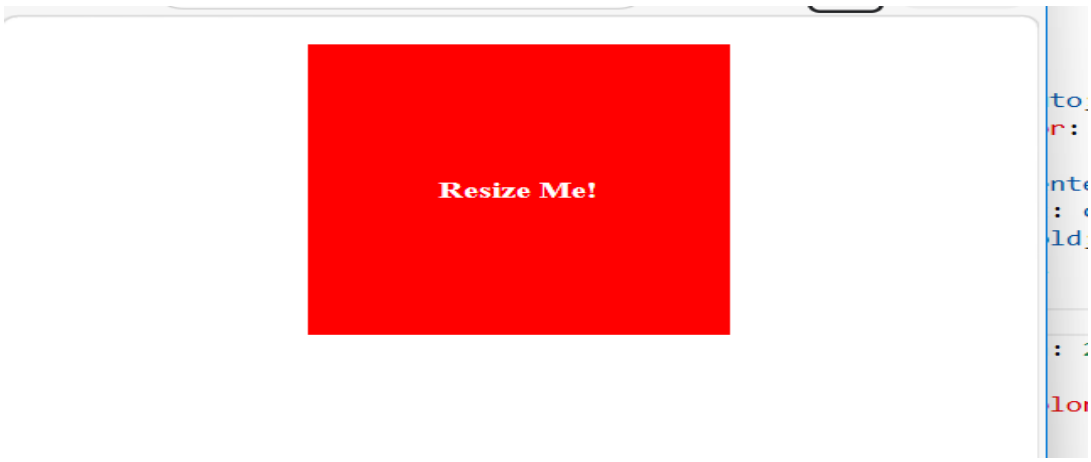
Center

```html
<!DOCTYPE html>
<html>
<head>
    <title>Flexbox Questions</title>
    <link rel="stylesheet" href="p6.css">
</head>
<body>

    <!-- Q1 Navbar -->
    <ul class="navbar">
        <li><a href="#">Home</a></li>
        <li><a href="#">About</a></li>
        <li><a href="#">Services</a></li>
        <li><a id="#">Contact</a></li>
    </ul>

    <!-- Q2 Centered Div -->
    <div class="outer">
        <div class="inner">Center</div>
    </div>



</body>
</html>
```

p6.css > ...

```css
.navbar {
    list-style: none;
    display: flex;
    justify-content: space-evenly;
    background-color: black;
    padding: 10px;
    margin: 0;
}
.navbar li a {
    color: white;
    text-decoration: none;
    font-size: 18px;
}
.outer {
    height: 300px;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: lightgray;
    margin-top: 50px;
}
.inner {
    height: 100px;
    width: 100px;
    background-color: teal;
    color: white;
    text-align: center;
    line-height: 100px;
}
```


Resize Me!

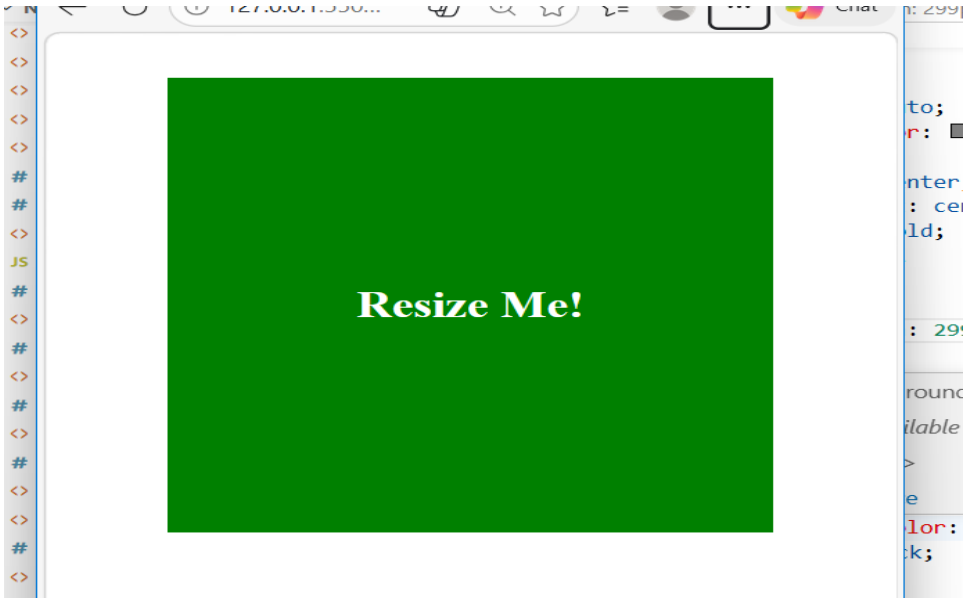

Resize Me!

7] Q1: Add a media query to implement the following:

- the color of a div changes to green for viewport width less than 300px.
- the color of a div changes to pink for width between 300px & 400px.
- the color of a div changes to red for width between 400px & 600px
- the color of a div changes to blue for width above 600px.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Media Query</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link rel="stylesheet" href="p7.css">
</head>
<body>

    <div class="box">Resize Screen</div>

</body>
</html>
```

127.0.0.1:550...
Chat
Resize Me!


Resize Me!

```
.box {
    width: 200px;
    height: 200px;
    margin: 20px auto;
    background-color: gray;
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: bold;
    color: white;
  }

  @media (max-width: 299px){
    .box {
      background-color: green;
    }
  }

  @media (min-width: 300px) and (max-width: 400px){
    .box {
      background-color: pink;
      color: black;
    }
  }

  @media (min-width: 401px) and (max-width: 600px){
    .box {
      background-color: red;
    }
  }

  @media (min-width: 601px){
    .box {
      background-color: blue;
    }
  }
```

output

8] Q1 : Create a simple loader using CSS

o Step1 : create a div with circular shape & a thick border from one end (top/bottom/left/right)

o Step2 : To make it spin create an animation which transforms it from 0deg to 360deg.

o Step3 : Add the animation property to the loader with infinite duration.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CSS Loader</title>
    <link rel="stylesheet" href="p8.css">
</head>
<body>

    <div class="loader"></div>

</body>
</html>
```

```css
body {
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
}

.loader {
    width: 80px;
    height: 80px;
    border: 10px solid lightgray;
    border-top: 10px solid blue; /* thick border on one side */
    border-radius: 50%;
}
@keyframes spin {
    from {
        transform: rotate(0deg);
    }
    to {
        transform: rotate(360deg);
    }
}
.loader {
    animation: spin 1s linear infinite;
}
```

Welcome

This is a simple paragraph.

Click Me

Hello!

Text changed using JavaScript.

Click Me

9] Q1: Create a simple HTML page with a heading and a paragraph and a button. Link a separate JavaScript file.

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>JS Linking</title>
</head>
<body>

    <h1 id="heading">Welcome</h1>
    <p id="para">This is a simple paragraph.</p>
    <button onclick="changeText()">Click Me</button>

    <!-- Link external JavaScript -->
    <script src="p9.js"></script>

</body>
</html>
```

```javascript
function changeText() {
    document.getElementById("heading").innerText = "Hello!";
    document.getElementById("para").innerText = "Text changed using JavaScript.";
}
```

Lion

Welcome to the Animal Page!

King of Jungle 🦁

Welcome to the Animal Page!

10] Q1: Create a simple webpage that displays the name of an animal at the top of the page. Add an interactive feature so that when the mouse pointer moves over the animal's name, the name changes to a fun nickname. When the mouse pointer moves away, the name should change back to the original animal name.

To make the page more interactive, also use set time out to display a small message after a few seconds. For example, after 3 or 4 seconds, a message such as "Welcome to the Animal Page!"

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Animal Page</title>
</head>
<body>

    <h1 id="animal">Lion</h1>

    <p id="message"></p>

    <script src="p10.js"></script>

</body>
</html>
```

```js
let animal = document.getElementById("animal");
animal.addEventListener("mouseover", function() {
    animal.innerText = "King of Jungle 🦁";
});

animal.addEventListener("mouseout", function() {
    animal.innerText = "Lion";
});

setTimeout(function() {
    document.getElementById("message").innerText = "Welcome to the Animal Page!";
}, 3000);
```

11] Q1: Create a webpage with a heading and enough content so the page can scroll. Add two interactive features using JavaScript.

1. Dark Mode

• Add a button to toggle dark and light mode.

• When clicked: o Background and text colors change. o Button text changes: 🌙 Dark Mode ↔ ☀️ Light Mode.

2. Smart Scroll-to-Top

• Add a “ ⬆️ Back to Top” button at the bottom corner of the page.

• Button appears only when the user scrolls down.

• Clicking the button smoothly scrolls the page to the top.

☀ Light Mode

Welcome to Nature World

Nature is the most beautiful and precious gift we have. From tall mountains to deep oceans, every part of nature teaches us something about balance and life. Spending time in nature helps reduce stress and improves mental health.

Forests are home to millions of animals and plants. They provide oxygen, food, and shelter. Protecting forests is essential for maintaining ecological balance and preventing climate change.

Water bodies like rivers, lakes, and oceans are equally important. They support marine life and provide water for drinking, farming, and industries. Clean water is necessary for all living beings.

We must take responsibility to protect our environment. Simple actions like planting trees, reducing plastic usage, and conserving water can make a big difference.

Let's work together to create a greener and healthier planet for future generations

🌙 Dark Mode

Welcome to Nature World

Nature is the most beautiful and precious gift we have. From tall mountains to deep oceans, every part of nature teaches us something about balance and life. Spending time in nature helps reduce stress and improves mental health.

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We must take responsibility to protect our environment. Simple actions like planting trees, reducing plastic usage, and conserving water can make a big difference.

Let's work together to create a greener and healthier planet for future generations

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Dark Mode & Scroll</title>
    <link rel="stylesheet" href="p11.css">
</head>
<body>
    <button id="modeBtn">🌙 Dark Mode</button>
    <h1>Welcome to Nature World </h1>

    <p>
        Nature is the most beautiful and precious gift we have. From tall mountains to deep oceans,
        every part of nature teaches us something about balance and life. Spending time in nature
        helps reduce stress and improves mental health.
    </p>

    <p>
        Forests are home to millions of animals and plants. They provide oxygen, food, and shelter.
        Protecting forests is essential for maintaining ecological balance and preventing climate change.
    </p>

    <p>
        Water bodies like rivers, lakes, and oceans are equally important. They support marine life
        and provide water for drinking, farming, and industries. Clean water is necessary for all living beings.
    </p>

```

```html
    <p>
        Water bodies like rivers, lakes, and oceans are equally important. They support marine life
        and provide water for drinking, farming, and industries. Clean water is necessary for all living beings.
    </p>

    <p>
        We must take responsibility to protect our environment. Simple actions like planting trees,
        reducing plastic usage, and conserving water can make a big difference.
    </p>

    <p>
        Let's work together to create a greener and healthier planet for future generations
    </p>

    <!-- Back to Top Button -->
    <button id="topBtn">↑ Back to Top</button>

    <script src="p11.js"></script>
</body>
</html>
```

```css
body {
    font-family: Arial, sans-serif;
    margin: 20px;
    transition: 0.3s;
}

/* Dark Mode */
.dark {
    background-color: black;
    color: white;
}

/* Mode Button */
#modeBtn {
    padding: 10px;
    margin-bottom: 20px;
}

/* Back to Top Button */
#topBtn {
    position: fixed;
    bottom: 20px;
    right: 20px;
    padding: 10px;
    display: none;
}
```

```javascript
let modeBtn = document.getElementById("modeBtn");
let topBtn = document.getElementById("topBtn");
modeBtn.addEventListener("click", function () {
    document.body.classList.toggle("dark");

    if (document.body.classList.contains("dark")) {
        modeBtn.innerText = "☀️ Light Mode";
    } else {
        modeBtn.innerText = "🌙 Dark Mode";
    }
});

window.addEventListener("scroll", function () {
    if (window.scrollY > 100) {
        topBtn.style.display = "block";
    } else {
        topBtn.style.display = "none";
    }
});
topBtn.addEventListener("click", function () {
    window.scrollTo({
        top: 0,
        behavior: "smooth"
    });
});
```